



Uveal melanoma

Early ultrasonographic tumor regression after linear accelerator stereotactic fractionated photon radiotherapy of choroidal melanoma as a predictor for metastatic spread

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ABSTRACT

Background and purpose: During extended follow-up (of up to 15 years), approximately fifty percent of patients with choroidal melanoma will develop metastatic disease and eventually die. Thus, continuing research on prognostic factors, early detection and treatment is necessary. Height regression rates both after plaque brachytherapy and proton beam irradiation have been shown to have prognostic value. The purpose of this study was to analyze the influence of early tumor regression rate after treatment of choroidal melanoma with LINAC stereotactic fractionated radiotherapy (SFRT) as an independent risk factor for metastasis.

Material and methods: 256 patients with choroidal melanoma treated with LINAC SFRT were included. Follow-up included standardized echography yielding apical height, smallest and largest basal linear diameter, tumor volume and mean reflectivity. The influence of baseline measurements and of a longitudinal, normalized area under the curve coefficient (NAC) of the latter marker on metastasis risk was assessed.

Results: NAC for tumor thickness at months 3, 6, and 12 had a statistically significant ($p < 0.001$) non-linear effect on risk of metastasis. Additionally, ultrasonographic baseline tumor dimensions, but not internal reflectivity were found to be statistically significant risk factors for metastasis.

Conclusions: Our results demonstrate a non-linear influence of regression rate of choroidal melanoma as independent risk factor of metastatic disease after LINAC SFRT. These prove the clinical experience that, in comparison to rather slow regressions, very quick and very slow early tumor responses to LINAC SFRT are associated with a significantly higher metastasis risk.

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Uveal melanoma is the most common primary intraocular malignancy in adults and can present in the choroid (90%), the ciliary body (6%) and in the iris (4%) [1]. Treatment options include plaque brachytherapy (mostly using iodine-125, ruthenium-106, and strontium-90), charged-particle proton beam radiation, linear accelerator (LINAC) based stereotactic fractionated radiotherapy (SFRT), gamma knife radiosurgery and enucleation [2–16]. It was shown that, concerning mortality rates, iodine-125 brachytherapy and primary enucleation did not differ for up to 12 years after treatment [17]. In contrast to single-fraction radiosurgery, stereotactic radiotherapy using LINAC offers the advantage of fractionated therapy resulting in reduced toxicity while providing tumor

control rates reported over 90%, 5 and 10 years after treatment [18–23]. At the Medical University of Vienna, SFRT is applied according to a standardized protocol if the initial height of the choroidal melanoma is 7 mm or higher, in case of juxtapapillary and/or juxtamacular location, and if the central tumor distance to the optic disc and/or the macula is less than 3 mm [22,24–26]. The articles published by our group provide an extensive overview about tumor control and radiation side effects of this treatment [18,22,27]. Although metastasis is rare at the time of initial ocular presentation, approximately 50% of patients develop metastatic disease and eventually die of melanoma [28]. Therefore, global research focuses on prognostic factors, early detection and treatment of metastatic disease. It is a well-known fact that the most significant prediction of melanoma-specific mortality is dependent on tumor specific genetic alterations [29–34]. Additionally, age, initial tumor dimensions and location, retinal detachment, as well

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